EPFL - Computer Science

COM-402 Information Security and Privacy

Lecture 01 — 08/09/2025

Author: Jonathan Dutoit

Introduction

This course is a broad overview class for security and privacy.

Exercices and Homeworks

An exercise sheet is distributed each week with both theoretical and practical questions about the lecture (including some coding exercises).

They are similar style as the exam questions. Solution sheet is distributed on Wednesday afternoon (5pm)

Quizzes

After each major block, we release a quiz with multiple choice questions about the homework and lectures.

Questions may be practical (i.e., homework related) or related to the lectures.

Recommendation: They recommend we do it whenever they come out and try to do a serious attempts.

Course Agenda

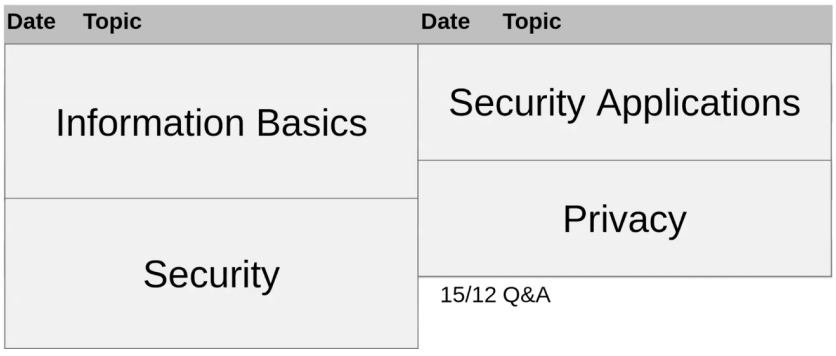


Figure 1: Course Agenda separeted by different themes

Information basics

- Cyber threats
- Crypto Basics
- Access Control

Security

- Data Security
- Programming Language Security and Compartments
- OWASP and Software Security

Security Applications

- Automated Testing / Fuzzing
- Mobile Security (Android as example)
- Network and Operations Security
- TEEs and Side Channels

Privacy

- Privacy
- Machine Learning Security and Privacy

Summary

Security and privacy is about controlling risk

You can't be secure without:

- knowing your assets, threats, and risks
- have some basic trust assumptions (e.g., encryption cannot be broken, my sysadmin is honest, etc.)

Solved by defining a policy, selecting mechanisms and implementing them (in practice much harder than it sounds)

Security is an **ongoing process**, requires regular audit/improvement